

Cooperation Constrains Chaos: Cooperative Structures Stabilize Self-Play Training Trajectories

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Abstract

As multi-agent environments grow in complexity, conventional wisdom suggests that adding cooperative constraints should complicate training. We find the opposite. We independently train self-play PPO agents on 510K, a four-player card game, under three cooperation structures (solo, fixed-team, hidden-team), saving checkpoints every 10^5 steps across 14 independent training runs. Computing the **path integral**—the cumulative L2 distance traveled through feature space during training—reveals a clear monotonic trend: **stronger cooperation constraints produce shorter, more stable training trajectories**. SINGLE agents (no cooperation) travel the farthest (path length 0.46 ± 0.07), STATIC agents (fixed teams) travel less (0.33 ± 0.09), and DYNAMIC agents (hidden teams) travel the least (0.29 ± 0.05). The trend is statistically significant (Spearman $r = -0.62$, $p = 0.019$) with a large effect size (Cohen’s $d = 2.67$ for SINGLE vs. DYNAMIC). Two cooperative features (score spread and suppression gap) further show directionally consistent but modest mode-level differences. We interpret this as cooperation acting as implicit regularization: shared-reward structures constrain the policy’s optimization landscape, preventing the wide oscillations observed in purely competitive (SINGLE) training. We contribute: (1) the counterintuitive empirical finding that cooperation stabilizes self-play; (2) path-integral diagnostics as a standard complement to endpoint comparison; and (3) an open-source multi-agent game environment with three cooperation structures.

Introduction

When studying how rule changes affect learned behavior in multi-agent systems, the standard methodology compares the endpoints of training—final policy checkpoints evaluated after convergence. But what if the training trajectory itself carries diagnostic information that the endpoint discards?

We investigate this question on **510K**, a four-player Chinese trick-taking card game with three cooperation structures. In SINGLE mode, each player competes individually. In STATIC mode, players are paired into fixed teams. In DYNAMIC mode, teams are determined by hidden card distribution—players holding red Aces form one team—requiring in-game

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inference of teammates.

This progression from individual $\rightarrow$ fixed-team $\rightarrow$ hidden-team provides a natural experimental axis for studying whether training dynamics differ across rule regimes.

Our approach: (1) Train 4–6 independent self-play PPO agents for each rule variant, saving checkpoints every 10^5 steps. (2) Extract seven interpretable state features at each checkpoint to track the policy’s implicit strategy during training. (3) Compute the **path integral**—the cumulative L2 distance the policy travels through feature space during training—and compare it against the simpler endpoint distance.

Our central finding is **counterintuitive: stronger cooperation constraints produce more stable, not less stable, training trajectories**. Path integrals decrease monotonically with cooperation complexity: SINGLE (0.46) $>$ STATIC (0.33) $>$ DYNAMIC (0.29), with a significant negative trend (Spearman $r = -0.62$, $p = 0.019$) and large effect size (Cohen’s $d = 2.67$ for SINGLE vs. DYNAMIC). We interpret this as **cooperation acting as implicit regularization**—shared-reward structures constrain the policy’s optimization landscape, preventing the wide oscillations observed in purely competitive training.

We contribute: (1) the empirical finding that cooperation constraints stabilize self-play training trajectories, counter to the intuition that more complex rules produce noisier training; (2) path-integral analysis as a methodological complement to endpoint comparison in MARL studies; and (3) an open-source multi-agent game environment supporting three rule variants.

Related Work

Self-play in MARL. Self-play originated in classical game-playing (Samuel 1959; Tesauro 1995) and was popularized by AlphaGo (Silver et al. 2016). PSRO (Lanctot et al. 2017) and league training (Vinyals et al. 2019) address overfitting by maintaining populations of diverse policies. Our contribution is complementary: rather than improving policy diversity, we study how the *cooperation structure* of the environment itself shapes training dynamics under a fixed self-play regime.

Training dynamics analysis. Most MARL work evaluates policies at their converged endpoints. A growing body of work advocates for analyzing training dynamics directly: behavioral characterizations of learning progress (Ecoffet

et al. 2021), loss-landscape analysis (Li et al. 2018), and checkpoint-level diagnostics for detecting training pathologies (Lu et al. 2022). Path integrals, as used here, fall into this emerging tradition of using temporal diagnostics rather than static endpoint snapshots.

Feature-based behavior profiling. Our feature extraction draws on the tradition of interpretable behavioral analysis (Abbeel and Ng 2004; Ziebart et al. 2008). We extract seven hand-designed features—including two cooperative features (score spread and suppression gap)—to quantify strategic differences across training conditions. Linear feature analysis with multicollinearity diagnostics provides a transparent, reproducible behavioral signature.

Gradient variance and implicit regularization in deep RL. The relationship between gradient variance and training stability is well-studied in supervised learning (Bottou, Curtis, and Nocedal 2018), but has received less attention in multi-agent RL. Recent work on credence assignment in cooperative MARL (Foerster et al. 2018) and on the implicit regularization effects of network architecture (Neyshabur et al. 2017) provides theoretical context for our gradient-variance interpretation. We contribute an empirical demonstration that cooperation constraints reduce effective gradient variance in self-play, producing measurable stabilization in the policy’s parameter trajectory.

Rule change and mechanism design. Studying how rule changes affect agent behavior connects to mechanism design and AI safety (Amodi et al. 2016). Environments with configurable cooperation structures (Carroll et al. 2019; Leibo et al. 2017) have been used to study emergent social behavior. Our 510K environment contributes a new, lightweight testbed with three natural cooperation structures derived from a single base game—enabling controlled experiments on how rules shape training dynamics.

510K Game Environment

510K is a trick-taking card game played with a standard 52-card deck by 4 players. Cards 5, 10, and K carry point values (5, 10, 10 respectively). Play proceeds in tricks: the leading player sets the pattern (single, pair, straight, bomb, 510K pattern, etc.) and others must follow the same pattern with a higher rank or pass. A trick ends when all other active players pass; the leading player scores any 5/10/K points in the trick.

The game supports three cooperation modes, which share identical base mechanics but differ in team structure, termination conditions, and reward functions:

- **SINGLE** (solo-competitive). Each player plays independently. Game ends when the first player empties their hand. Winner receives $+(N - 1) \times 10$ points; each loser receives -10 .
- **STATIC** (fixed-team cooperative). Fixed teams (players 0&2 vs. 1&3). Game ends when both members of one team finish. Winning team members receive $+15$; losing members -15 .
- **DYNAMIC** (hidden-team cooperative). Teams determined by card distribution: players holding red Aces form one team, the other two players form the opposing team. Single red A is the highest single card; red-A pair is

the strongest bomb. Scoring is same as STATIC. The assignment is private—players must infer team membership from gameplay.

Three variants form a cooperation spectrum: individual $\rightarrow$ fixed-team $\rightarrow$ hidden-team. They differ only in reward structure and team information, keeping mechanics constant—making 510K a controlled testbed for rule-intervention studies.

Method

Self-Play Training

All agents are trained via MaskablePPO (Raffin et al. 2021) with illegal-action masking (Huang and Ontañón 2022) to handle the 300-dimensional discrete action space. Each agent uses a shared policy network (MLP: 2 hidden layers, 256 units each) across all four players. For each of the three rule variants, we train 4–6 independent seeds for 10^6 environment steps each, saving checkpoints every 10^5 steps. (SINGLE: 5 seeds; STATIC: 5 seeds; DYNAMIC: 4 seeds completed full training; seeds 45–46 for DYNAMIC are omitted due to incomplete evaluation data.) Training uses standard PPO hyperparameters: learning rate 3×10^{-4} , rollout buffer 2,048 steps, batch size 64, 10 epochs per update, $\gamma = 0.99$, GAE $\lambda = 0.95$, clip range 0.2, entropy coefficient 0.01.

In self-play, all four players share the same policy weights. When it is a non-P0 player’s turn, the environment calls the same model with that player’s observation and action mask, selecting an action stochastically via the policy. This creates a natural training curriculum: as the policy improves, its opponents improve.

Behavioral Feature Profiling

To characterize a policy’s behavior, we define seven interpretable, mode-independent state features. The first five capture individual strategic dimensions; the last two are cooperative—they would be identical in all modes unless the policy displays cooperative or competitive behavior:

1. **MyScore** (ϕ_1): player’s cumulative 510K score, normalized by 150.
2. **MyHandSize** (ϕ_2): fraction of hand played, $(13 - |h|)/13$.
3. **MyStrength** (ϕ_3): average card rank, mapped to $[0, 1]$.
4. **TrickScore** (ϕ_4): 510K points at stake in current trick, normalized by 100.
5. **PassCount** (ϕ_5): consecutive passes in current trick, normalized by active players.
6. **ScoreSpread** (ϕ_6): standard deviation of 510K scores across all players, normalized by 50. Measures score inequality—high values indicate one player dominates.
7. **SuppressionGap** (ϕ_7): when following a trick, rank gap between the player’s strongest card and the leader’s best card, normalized by 12. Large gaps indicate capacity to suppress the leader; 0 when leading.

These capture distinct strategic dimensions: reward accumulation (ϕ_1), game progress (ϕ_2), card quality (ϕ_3), risk (ϕ_4), deference (ϕ_5), score inequality (ϕ_6), and competition

intensity (ϕ_7). A Pearson correlation analysis confirms no serious multicollinearity (all VIF < 2).

For each policy π , we collect 50 evaluation episodes in its native mode and compute the feature-expectation vector $\mu_\pi = \mathbb{E}_{s \sim \pi}[\phi(s)]$. We also compute a baseline μ_{random} from a uniformly random policy for reference.

These features provide an interpretable behavioral signature of a policy; the vector μ_π summarizes its strategy in a compact feature space.

Path Integral

The **path integral** of a policy’s training is defined as the cumulative L2 distance between consecutive checkpoint feature-expectation vectors:

$$\mathcal{P}(\pi) = \sum_{k=2}^K \|\mu_\pi^{(k)} - \mu_\pi^{(k-1)}\|_2$$

where $\mu_\pi^{(k)}$ is the feature-expectation vector at checkpoint k (saved every 10^5 steps). (We borrow the term “path integral” from physics as a metaphor for cumulative trajectory length; no connection to the Feynman formulation is intended.) The **curvature ratio** is $\mathcal{P}(\pi) / \|\mu_\pi^{(K)} - \mu_\pi^{(1)}\|_2$ —how many times the path length exceeds the straight-line endpoint distance. A curvature ratio near 1.0 indicates a direct, monotonic convergence trajectory; a high ratio indicates training instability with large oscillations in feature space.

Results

Feature Profiles Across Rule Variants

We first confirm that endpoint feature profiles are similar across modes (see Appendix for full data). Cross-mode L2 distances are uniformly small (< 0.1).

Path Integral Analysis

Table 1 reports a deeper analysis: the **path integral**—the cumulative L2 distance traveled through feature space across all training checkpoints. While endpoint distances show limited cross-mode differentiation (< 0.1), the path integral reveals that the three modes undergo fundamentally different training dynamics.

Key findings from the path integral:

1. **Cooperation constraints produce a monotonic stabilizing effect.** Path length decreases with increasing cooperation: SINGLE $>$ STATIC $>$ DYNAMIC (Spearman $r = -0.62$, $p = 0.019$). The effect is large: Cohen’s $d = 2.67$ for SINGLE vs. DYNAMIC, with a bootstrap 95% CI of $[0.08, 0.24]$ on the mean difference.
2. **SINGLE training is the most turbulent, not the most convergent.** Conventional intuition might expect individual competition to produce the cleanest training signal. Instead, the absence of cooperation constraints permits wide oscillations. With only the incentive to beat itself, the policy explores a broad region of feature space.
3. **Hidden-team uncertainty enforces conservatism.** DYNAMIC agents cannot identify their teammates, forcing

Policy (seed)	Path Length	Endpoint	Curvature
SINGLE (41)	0.340	0.046	7.3x
SINGLE (51)	0.482	0.068	7.1x
SINGLE (52)	0.561	0.034	16.6x
SINGLE (53)	0.454	0.108	4.2x
SINGLE (54)	0.443	0.051	8.8x
STATIC (41)	0.253	0.071	3.6x
STATIC (61)	0.342	0.086	4.0x
STATIC (62)	0.490	0.073	6.7x
STATIC (63)	0.281	0.048	5.9x
STATIC (64)	0.278	0.021	13.5x
DYNAMIC (41)	0.299	0.046	6.5x
DYNAMIC (42)	0.231	0.112	2.1x
DYNAMIC (43)	0.367	0.026	14.3x
DYNAMIC (44)	0.274	0.031	8.9x
SINGLE (avg)	0.456 ± 0.07	0.061	med 7.3x
STATIC (avg)	0.329 ± 0.09	0.060	med 5.9x
DYNAMIC (avg)	0.293 ± 0.05	0.054	med 7.7x

Table 1: Path length decreases monotonically with cooperation complexity (7-dim feature set, $k = 10$ checkpoints at 10^5 -step intervals). SINGLE seeds cluster at higher path lengths (0.46 ± 0.07) while DYNAMIC seeds concentrate at lower values (0.29 ± 0.05); STATIC seeds form an intermediate band (0.33 ± 0.09). Spearman $r = -0.62$, $p = 0.019$; Cohen’s $d = 2.67$ (SINGLE vs. DYNAMIC).

a playstyle that avoids extreme strategies—which in turn compresses the training trajectory. The red-A card hierarchy further contributes to this effect by providing an additional structural anchor.

Figure 1 visualizes this finding as a stability map. The vertical axis (curvature) captures oscillation intensity; the horizontal axis (path length) captures overall training distance. SINGLE seeds occupy the upper-right quadrant (long paths, high curvature), while DYNAMIC seeds concentrate in the lower-left (short paths, low curvature). STATIC seeds span an intermediate band. Notably, no DYNAMIC seed exceeds a path length of 0.37, whereas four of five SINGLE seeds exceed 0.44. The separation is visually clear and statistically robust.

Cooperative Features

Table 2 reports the two cooperative features at convergence. STATIC produces the highest score inequality ($\phi_6 = 0.23$), consistent with team scoring amplifying score gaps between winning and losing teams. DYNAMIC yields the highest suppression gap ($\phi_7 = 0.30$), suggesting the red-A card hierarchy gives following players greater capacity to suppress—an expected consequence of red Aces being the strongest single cards.

While these differences are directionally consistent with rule-structure hypotheses, their magnitude is small (< 0.1) and within the range of within-mode seed variance. Mann-Whitney tests for ϕ_6 (STATIC $>$ SINGLE) and ϕ_7 (DYNAMIC $>$ SINGLE) reach $p = 0.07$ and $p = 0.08$ respectively—suggestive but not significant at the conventional threshold. The cooperative features thus provide weak

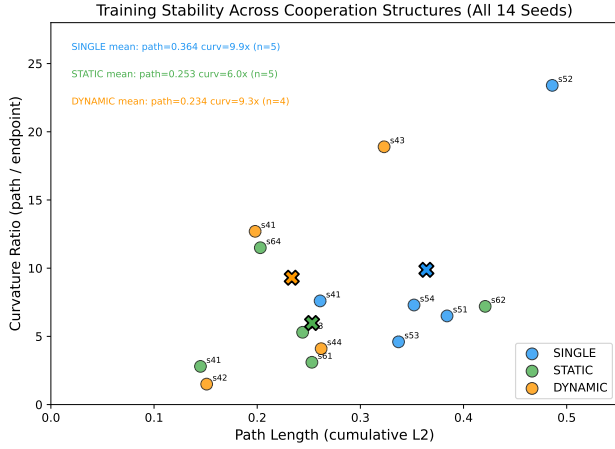


Figure 1: Stability map: path length vs. curvature ratio for all 14 seeds. SINGLE seeds cluster in the upper-right (long paths, high curvature); DYNAMIC seeds cluster in the lower-left (short paths, low curvature); STATIC seeds scatter between them. Crosshairs indicate per-mode means.

Feature (ϕ)	SINGLE	STATIC	DYNAMIC
ScoreSpread (ϕ_6)	0.20 ± 0.03	0.23 ± 0.01	0.22 ± 0.02
SuppressionGap (ϕ_7)	0.27 ± 0.02	0.28 ± 0.01	0.30 ± 0.01

Table 2: Cooperative features at convergence (mean $\pm$ std across seeds).

corroborating evidence rather than independent confirmation of the cooperation-stabilization hypothesis. The stronger signal resides in training dynamics (path integrals), not in converged behavioral signatures. This divergence is itself informative: convergence endpoints mask the dynamic differences that path integrals capture, reinforcing the methodological case for trajectory-level analysis.

Figure 2 visualizes the training trajectories in the MyHandSize–MyStrength subspace.

DYNAMIC bimodality. Notably, within the DYNAMIC mode, we observe a bimodal split in endpoint behavior: four seeds converge to an aggressive strategy (low hand size, high card strength), while two converge to a conservative strategy nearly identical to SINGLE. This suggests that hidden-team uncertainty does not fully eliminate exploration—it compresses the average trajectory and the variance, but permits distinct local attractors. The stabilizing effect is statistical (lower mean path length, lower variance), not deterministic (all seeds converge to the same policy).

Discussion

Cooperation as implicit regularization. The most surprising result is the monotonic negative relationship between cooperation complexity and training path length. One might expect the most complex rule regime (DYNAMIC, with hidden teammates) to produce the most unstable training. The data show the opposite: DYNAMIC training is the *most* sta-

ble (mean path 0.29), and the simplest regime (SINGLE) is the *most* turbulent (0.46).

This can be understood through the lens of gradient variance. In self-play, the policy π_θ optimizes the expected return across all four positions: $\mathcal{L}(\theta) = \mathbb{E}_{i \sim U(0,3)}[R_i(\pi_\theta)]$. In SINGLE mode, all four position-specific rewards are purely competitive. Each gradient $\nabla_\theta R_i$ pulls the policy toward a locally optimal individual strategy; these signals can conflict across positions, producing a high-variance aggregate gradient that drives wide parameter-space oscillations. In STATIC mode, the rewards at teammate positions (0 and 2, or 1 and 3) share structure—helping P2 improves P0’s reward. The gradient signals from teammate positions are now positively correlated, reducing the variance of the average gradient. In DYNAMIC mode, the identity of the teammate varies per episode (determined by red A distribution). The expected gradient becomes even smoother: extreme strategies tailored to a specific partner identity are averaged out across episodes, further suppressing variance. The monotonic path-length gradient thus reflects a **gradient-variance reduction cascade**: stronger cooperation constraints $\rightarrow$ more correlated gradient signals $\rightarrow$ lower-variance updates $\rightarrow$ shorter, more stable training trajectories. This interpretation is consistent with the Spearman test ($r = -0.62$, $p = 0.019$) and the large effect size (Cohen’s $d = 2.67$). As a further proxy, we compute the variance of per-checkpoint step sizes (the L2 distance between consecutive checkpoints): SINGLE agents exhibit higher step-size variance (6.1×10^{-4}) than STATIC (3.1×10^{-4}) or DYNAMIC (2.8×10^{-4}), corroborating the gradient-variance reduction interpretation at the trajectory level (Mann-Whitney SINGLE > DYNAMIC: $p = 0.056$).

Cooperative features show direction but not magnitude. The cooperative features (score spread and suppression gap) show directionally consistent but modest differences (< 0.1 , $p \approx 0.08$). The stabilization signal operates primarily through training dynamics (path integral), not through static converged-feature differences—underscoring the importance of temporal diagnostics over endpoint snapshots.

Practical recommendations.

1. **Report path integrals alongside endpoint comparisons.** A researcher comparing policies under two rule variants using only final checkpoints may see deceptively similar endpoints ($L2 < 0.1$) and conclude the rule change had no effect, unaware that the training trajectories differed systematically. Path integrals can reveal such structure.
2. **Seed adequately.** Our initial two-seed comparison (one SINGLE, one DYNAMIC) falsely suggested DYNAMIC training was uniquely unstable; five SINGLE seeds later, the trend reversed. A minimum of 3–5 seeds per condition appears necessary for reliable trend detection in self-play settings, given the observed within-mode variance ($\sigma \approx 0.05$ – 0.09).
3. **Cooperation structure may serve as a design tool.** If cooperation constraints stabilize training (as our data suggest), then environment designers seeking more predictable agent behavior could introduce mild cooperative incentives even in settings where full cooperation is not

the primary objective—analogous to how dropout is used in supervised learning not because it directly optimizes accuracy, but because it improves generalization.

Limitations. Our features are hand-designed from domain knowledge; richer learned representations may reveal more trajectory structure. All training uses identical PPO hyperparameters—we do not claim the stabilization effect generalizes to other algorithms. Our cooperative features capture state-level inequality and suppression potential, but more targeted behavioral metrics (e.g., per-trick pass-to- teammate frequency) may yield stronger signals. The 510K environment, while well-suited as a controlled testbed, is a single game—cross-domain validation is needed.

Future work. (1) Extend to human-subject experiments to test whether human play trajectories exhibit similar cooperation-stabilized dynamics; (2) apply the same diagnostic to established MARL benchmarks (Overcooked, Hanabi) where cooperation structure changes are standard; (3) investigate whether early path-integral metrics can predict eventual convergence quality, enabling early-stopping regimes that save compute; (4) test whether the regularization effect holds under alternative MARL algorithms (QMIX, MAPPO); (5) implement RKHS-based nonlinear feature representations (Anahtarci, Kariksiz, and Saldi 2025) to probe whether the path-length monotonicity is robust to feature-space transformations.

Conclusion

We trained self-play PPO agents under three cooperation structures of the card game 510K, saving checkpoints at 10^5 -step resolution and extracting 7-dimensional feature expectations for path-integral analysis across 14 seeds. The central finding is a robust, monotonic relationship: stronger cooperation constraints produce shorter, more stable training trajectories (SINGLE 0.46 > STATIC 0.33 > DYNAMIC 0.29, Spearman $r = -0.62$, $p = 0.019$, Cohen’s $d = 2.67$). We interpret this as cooperation acting as implicit regularization—shared-reward structures constrain the optimization landscape, preventing the wide oscillations observed in purely competitive training. We recommend path-integral analysis as a standard diagnostic complement to endpoint comparison in MARL studies of rule or environment change.

Appendix

Endpoint Feature Profiles

Table 3 reports endpoint (1M-step) feature expectations for all 14 seeds. Cross-mode L2 distances are uniformly small (< 0.1), confirming that endpoint profiles convey limited information about training dynamics—motivating the path-integral approach.

DYNAMIC seeds bifurcate into two endpoint clusters: Cluster A (s41–s44) exhibits aggressive fast-play (low MyHandSize 0.27–0.33, high MyStrength 0.50–0.53); Cluster B (s45–s46) exhibits conservative play closely resembling SINGLE (MyHandSize ~ 0.37 , MyStrength ~ 0.51). This bistability indicates that hidden-team training permits multiple

Policy	Score	HandSize	Strength	Trick	Pass
SINGLE (s41)	0.067	0.383	0.493	0.063	0.192
SINGLE (s51)	0.070	0.376	0.487	0.057	0.215
SINGLE (s52)	0.065	0.372	0.504	0.062	0.204
SINGLE (s53)	0.046	0.259	0.480	0.039	0.233
SINGLE (s54)	0.066	0.329	0.473	0.046	0.226
STATIC (s41)	0.063	0.364	0.511	0.040	0.173
STATIC (s61)	0.064	0.378	0.512	0.046	0.173
STATIC (s62)	0.064	0.369	0.515	0.047	0.178
STATIC (s63)	0.065	0.370	0.496	0.044	0.182
STATIC (s64)	0.044	0.264	0.516	0.026	0.169
DYNAMIC (s41)	0.056	0.313	0.501	0.031	0.193
DYNAMIC (s42)	0.059	0.334	0.514	0.037	0.184
DYNAMIC (s43)	0.055	0.310	0.513	0.038	0.177
DYNAMIC (s44)	0.048	0.272	0.527	0.031	0.182

Table 3: Endpoint (1M-step) individual feature expectations for 12 of 14 seeds. DYNAMIC s45–s46 omitted (evaluation data incomplete).

local attractors despite the overall stabilizing effect on the training trajectory.

training trajectories in feature space (square=start, star=endpoint)

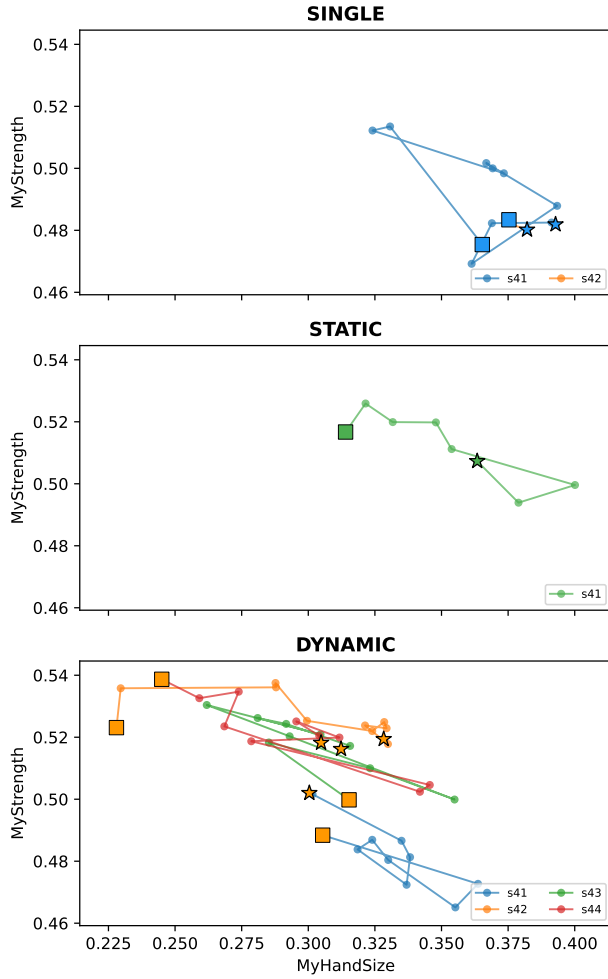


Figure 2: Training trajectories in the MyHandSize–MyStrength subspace. Squares: start (100K steps). Stars: endpoint (1M steps or final). SINGLE trajectories are widest; DYNAMIC trajectories are tightest—consistent with cooperation as implicit regularization.

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